

Anna University

MID-TERM EXAMINATION ? 2025

Material Science and Metallurgy ? Mid-Term Examination, 2025

Subject Code: ME303 Semester: 3 Regulation: R23

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Crystal Structure and Imperfections)

1. Describe the working principle of Crystal Structure and Imperfections with necessary equations. [12 Marks]
2. Write short notes on Crystal Structure and Imperfections. [12 Marks]
3. Compare and contrast the key concepts involved in Crystal Structure and Imperfections. [12 Marks]

SECTION B (Unit 2: Phase Diagrams)

4. Compare and contrast the key concepts involved in Phase Diagrams. [8 Marks]
5. Discuss the engineering applications of Phase Diagrams. [12 Marks]
6. Solve a problem related to Phase Diagrams. Show all the steps clearly. [8 Marks]

SECTION C (Unit 3: Heat Treatment of Steels)

7. Write short notes on Heat Treatment of Steels. [12 Marks]
8. Solve a problem related to Heat Treatment of Steels. Show all the steps clearly. [12 Marks]
9. List the advantages and limitations of Heat Treatment of Steels. Explain any two in detail. [12 Marks]

SECTION D (Unit 4: Mechanical Properties of Materials)

10. Compare and contrast the key concepts involved in Mechanical Properties of Materials. [8 Marks]
11. How is Mechanical Properties of Materials implemented? Give an algorithm / procedure and explain. [12 Marks]
12. Discuss the engineering applications of Mechanical Properties of Materials. [12 Marks]

SECTION E (Unit 5: Non-Ferrous Materials and Alloys)

13. Discuss the engineering applications of Non-Ferrous Materials and Alloys. [12 Marks]
14. Write short notes on Non-Ferrous Materials and Alloys. [8 Marks]
15. Write short notes on Non-Ferrous Materials and Alloys. [12 Marks]

***** End of Question Paper *****